ODE-6

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n-th Second Order linear homogeneous eqn.

Considering,

$$\left(a_0\frac{d^ny}{dx^n} + a_1\frac{d^{n-1}y}{dx^{n-1}} + a_2\frac{d^{n-2}y}{dx^{n-2}} + \cdots + a_{n-1}\frac{dy}{dx} + a_n\right)y = f(x)$$

→ a non-homogeneous linear DE with constant-coefficients of order n, $0 \neq a_1, a_2, \ldots a_n$ are constants

→ Symbolically, $(a_0D^n + a_1D^{n-1} + a_2D^{n-2} + \cdots + a_{n-1}D + a_n)y = f(x)$

where $D \equiv \frac{d}{dx}$

→ $\Rightarrow Ly = f(x)$, $L \equiv a_0D^n + a_1D^{n-1} + \cdots + a_{n-1}D + a_n$ is a linear operator

Auxiliary eqn. or characteristic eqn. is

$$a_0\lambda^n + a_1\lambda^{n-1} + a_2\lambda^{n-2} + \cdots + a_{n-1}\lambda + a_n = 0 \quad \text{——(1)}$$

Case-I Roots of eqn (1) are real and distinct,

say, $\lambda = \lambda_1, \lambda_2, \ldots \lambda_n$,

then general soln is: $y = c_1e^{\lambda_1 x} + c_2e^{\lambda_2 x} + \cdots + c_ne^{\lambda_n x}$

$c_1, c_2 \ldots c_n$ are arbitrary constants

Case-II Roots of eqn (1) are real and repeated, say,

$\lambda = \lambda_*, \lambda_*, \lambda_3, \lambda_4, \ldots \lambda_n$

then $y = (c_1 + c_2x)e^{\lambda_* x} + c_3e^{\lambda_3 x} + c_4e^{\lambda_4 x} + \cdots + c_ne^{\lambda_n x}$

Case-III Roots of eqn. (1) are imaginary and distinct,

say, $\lambda = \alpha \pm i\beta, \lambda_3, \lambda_4, \ldots \lambda_n$ (α_i's are real and have conjugate pairs)

G.S $y = e^{\alpha x}[c_1\cos\beta x + c_2\sin\beta x] + c_3e^{\lambda_3 x} + c_4e^{\lambda_4 x} + \cdots + c_ne^{\lambda_n x}$

$\therefore y = c_1e^{(\alpha+i\beta)x} + c_2e^{(\alpha-i\beta)x} + c_3e^{\lambda_3 x} + \cdots + c_ne^{\lambda_n x}$

Case-IV If imaginary roots are repeated,

$\lambda = \alpha \pm i\beta, \alpha \pm i\beta, \lambda_5, \lambda_6, \ldots \lambda_n$

G.S $\Rightarrow y = e^{\alpha x}[(c_1 + c_2x)\cos\beta x + (c_3 + c_4x)\sin\beta x] + c_5e^{\lambda_5 x} + \cdots + c_ne^{\lambda_n x}$

Second order linear homogeneous eqn.

$$a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + c = 0 \quad \text{— (1)}$$

Trial soln: $y = e^{rx}$

Using it in (1), we get auxiliary/characteristic eqn.

$$ar^2 + br + c = 0$$

Solve for roots

(1) Real and distinct roots r_1, r_2: r_1x , r_2x

Solve for roots

① Real and distinct roots r_1, r_2:

$$y = c_1 e^{r_1 x} + c_2 e^{r_2 x}$$

② Real and repeated roots r_1, r_1:

$$y = (c_1 + c_2 x) e^{r_1 x}$$

③ Complex conjugate roots $\alpha \pm i\beta$:

$$y = e^{\alpha x} (c_1 \cos\beta + c_2 \sin\beta)$$

Examples

① $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6 = 0$

$(D^2 - 5D + 6)y = 0$, $\frac{d}{dx} \equiv D$

Trial soln: $e^{rx} = y$

Putting it we get auxiliary eqn.,

$r^2 - 5r + 6 = 0$

$r = 2, 3$

$y = c_1 e^{2x} + c_2 e^{3x}$

② $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4 = 0$

Aux. eqn: $r^2 - 4r + 4 = 0$

$\Rightarrow (r-2)^2 = 0$

$\Rightarrow r = 2, 2$ (real, repeated)

$y = (c_1 + c_2 x) e^{2x}$

③ $\frac{d^2y}{dx^2} + 4y = 0$

Aux. eqn: $r^2 + 4 = 0 \Rightarrow r = \pm 2i$ (complex conjugate)

$y = e^{0x} [c_1 \cos 2x + c_2 \sin x]$

$= c_1 \cos 2x + c_2 \sin x$

④ $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 5 = 0$

Aux. eqn: $r^2 + 2r + 5 = 0$

$r = \frac{-2 \pm \sqrt{4 - 20}}{2} = \frac{-2 \pm 4i}{2} = -1 \pm 2i$

$y = e^{-x} [c_1 \cos 2x + c_2 \sin 2x]$

⑤ $y'' + 6y' + 9 = 0$

aux. eqn → $r^2 + 6r + 9 = 0 \Rightarrow (r+3)^2 = 0 \Rightarrow r = -3, -3$

$y = (c_1 + c_2 x) e^{-3x}$

(6) $y''' + 6y'' + 11y' + 6y = 0$

aux. eqn $\Rightarrow$ $m^3 + 6m^2 + 11m + 6 = 0$

$\Rightarrow m^3 + m^2 + 5m^2 + 5m + 6m + 6 = 0$

$\Rightarrow m^2(m+1) + 5m(m+1) + 6(m+1) = 0$

$\Rightarrow (m+1)(m+2)(m+3) = 0$

C.F $\rightarrow c_1 e^{-x} + c_2 e^{-2x} + c_3 e^{-3x}$